SOLUTION
SECTION-I

Single Correct Answer Type **13 Q. [3 M (–1)]**

1. Ans. (C)

2. Ans. (B)

3. Ans. (C)

4. Ans. (B)

Sol. For an EM wave, direction of wave propagation is given by $\vec{E}\times\vec{B}$ & also $\vec{E}$ & $\vec{B}$ are mutually pependicular & in phase.

5. Ans. (B)

Sol. $I=\frac{I}{2}C\in_0 E_0^2$

$=\frac{1}{2}\times3\times10^8\times8.85\times10^{-12}\times1296=1.72\ W/m^2$

6. Ans. (C)

Sol. Average energy density of electric field

$=\frac{1}{2}\varepsilon_0E_{rms}^2$

$=\frac{1}{2}\varepsilon_0\left[\frac{E_0}{\sqrt{2}}\right]^2=\frac{1}{4}\varepsilon_0E_0^2=\frac{1}{4}\times8.8\times10^{-12}\times4^2$

$=35.2\times10^{-12}\ J/m^3$

7. Ans. (B)

Sol. $J_c=\sigma E_0\sin(\omega t-kx)$

$i_d=\in\in_0\frac{d\phi_E}{dt}=\in\in_0 A\frac{dE}{dt}$

$=\in\in_0\times A\,E_0\omega\cos(\omega t-kx)$

$Jd=\in\in_0E_0\omega$

$\frac{J_c}{J_d}=\frac{\sigma}{\in\in_0\omega}$

8. Ans. (B)

Sol. $\vec{E}=10\hat{j}\cos(6x+8z)=10\hat{j}\cos\left[\left(6\hat{i}+8\hat{k}\right).\left(x\hat{i}+z\hat{k}\right)\right]$

$\therefore\ \vec{k}=6\hat{i}+8\hat{k}$: This is the direction of 'c'.

$\vec{c}\times\vec{E}$ is direction of $\vec{B}:\left(6\hat{i}+8\hat{k}\right)\times\hat{j}=6\hat{k}-8\hat{i}$

$\therefore\vec{B}=\left(\frac{6\hat{k}-8\hat{i}}{c}\right)\cos\left[6x+8z-10ct\right]$ as $B=\frac{E}{c}$

9. Ans. (B)

Sol. $I=\frac{P}{4\pi r^2}=\frac{E_{rms}^2}{c\mu_0}$

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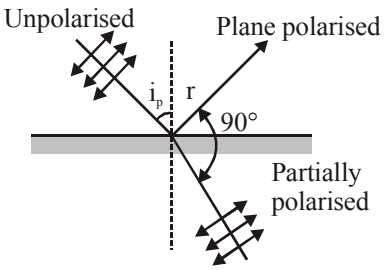
$$E_{rms} = \sqrt{\frac{250 \times 3 \times 10^8 \times 4\pi \times 10^{-7}}{4\pi(0.9)^2}} = 96 V/m$$

10. Ans. (B)

Sol. Polarization by scattering.

11. Ans. (C)

Sol. From Brewster's law reflected ray is perpendicular to refracted ray.



The reflected ray os obtained is plane polarized having is electric vector in the plane of incidence.

12. Ans. (A)

Sol. Resolving power is given by the receprocal of the minimum separation between two points seen as distinct

$$= \frac{D}{1.22\lambda L}$$

13. Ans. (B)

Sol. $A = A_0\left(\frac{\sin\beta}{\beta}\right)$

or $\frac{A}{A_0} = \frac{\sin\beta}{\beta}$

Now $\beta = \frac{\pi a \sin\theta}{\lambda} = \frac{\pi}{2}$

$\therefore \quad \frac{A}{A_0} = \frac{\sin\left(\pi/2\right)}{\left(\pi/2\right)} = \frac{2}{\pi}$

Multiple Correct Answer Type **3 Q. [4 M (–1)]**

14. Ans. (A,B,C)

15. Ans. (B,D)

16. Ans. (A,D)

Linked Comprehension Type (1 Para × 2Q.) [3 M (-1)]

(Single Correct Answer Type)

17. Ans. (B)

18. Ans. (A)

Subjective Type **11 Q. [4 M (0)]**

19. Ans. $\frac{\lambda^2 v}{4\pi^2 \varepsilon_0 a^2}\hat{k}$

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20. **Ans. (i)** $\frac{2\pi}{\lambda^2} I_0 \ln\left(\frac{a}{s}\right)\sin(2\pi vt)\hat{k}$ **, (ii)** $+\left(\frac{a}{2}\right)^2\left(\frac{2\pi}{\lambda}\right)^2 I_0 \sin 2\pi vt$ **, (iii)** $\frac{I_0^d}{I_0} = \left(\frac{a\pi}{\lambda}\right)^2$

21. **Ans. (a) 46 mA/m, (b)** $\sqrt{3} \times 10^8$ **m/s**

22. **Ans. 1 mJ**

23. **Ans. (a)** $\frac{r\varepsilon_0 u^2}{2\tau^2 d^2} t$ **, (b)** $\frac{\varepsilon_0 u^2 \pi r^2}{2d}$

24. **Ans. (a)** $\frac{\mu_0 n^2 I^2 r}{2\tau^2} t$ **; (b)** $\frac{\mu_0 n^2 I^2 \pi r^2 \ell}{2}$

25. **Ans. (a) 18.84 V/m; (b)** 1.57×10^{-9}

26. **Ans. 6.17**

27. **Ans. 1026.94**

28. **Ans. 50**

29. **Ans. (a) 0.01 A; (b) 2 s**

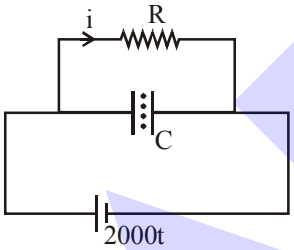
Sol. (a) $I_d = \varepsilon_0 \frac{d\phi_E}{dt} = \frac{\varepsilon_0 Ad\left(\frac{V}{d}\right)}{dt} = \frac{\varepsilon_0 A}{d}\frac{dq}{dt}$

$= C\frac{dv}{dt} = 5\times10^{-6}\times2000 = 0.01A$

(b) $iR = 2000\ t$

$0.01 \times 4 \times 10^5 = 2000\ t$

$\therefore\ t = 2s$



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